QUANT LAB

Integrated Counterparty Risk, XVA & Capital Analytics Platform

Document type	Project Requirements Document (PRD)
Version	1.0
Project classification	Quantitative Finance / Quantitative Development
Status	Proposed

Project vision

Build a reproducible, end-to-end derivatives counterparty risk platform that connects trade-level economics to portfolio-level valuation, exposure, capital and risk attribution.

Prepared as a functional specification for the design and implementation of Quant Lab.

Document structure

- 1. Project Overview
- 2. High-Level System Architecture
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- 9. Quantitative Validation Framework
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End-to-end processing flow

Portfolio Input	**Market Simulation**	**Pricing**	**Exposure**	**Netting & Collateral**
XVA	**Capital & RWA**	**Allocation**	**Reporting**	

1. Project Overview

Quant Lab is an integrated quantitative analytics platform designed to simulate, price, measure and attribute counterparty credit risk across a portfolio of derivative instruments. The platform accepts a synthetic portfolio, simulates underlying market states, calculates instrument valuations and generates future exposure profiles. Exposures are adjusted for netting and collateral arrangements before calculating counterparty risk metrics, valuation adjustments, regulatory capital and risk-weighted assets (RWA). Finally, portfolio-level metrics are attributed to individual trades using quantitative allocation methodologies.

1.1 Objectives

- Demonstrate derivative pricing and market risk factor modelling.
- Implement Monte Carlo simulation and exposure modelling.
- Build netting-set aggregation and collateral modelling capabilities.
- Measure counterparty credit risk and valuation adjustments (XVA).
- Calculate regulatory capital and RWA using documented methodologies.
- Attribute portfolio risk and capital to trades, with emphasis on Euler and Aumann-Shapley allocation.
- Demonstrate production-quality Python development and quantitative software architecture.

1.2 Design principles

- **Modularity:** Each quantitative component operates as an independent, reusable module.
- **Extensibility:** New products, risk factors, pricing models and regulatory methodologies can be added without redesigning the platform.
- **Traceability:** Portfolio metrics can be traced to source trades, scenarios and assumptions.
- **Reproducibility:** Identical inputs, configurations and random seeds produce identical results.
- **Quantitative integrity:** Calculations are supported by mathematical definitions, validation tests and reconciliation checks.

2. High-Level System Architecture

The platform follows a sequential processing architecture: **Input** → **Market Simulation** → **Pricing** → **Exposure** → **Netting & Collateral** → **XVA** → **Capital & RWA** → **Allocation** → **Reporting.**

2.1 Portfolio input and configuration

Ingest the synthetic portfolio and supporting assumptions.

- Trade dataset and product attributes
- Counterparty and netting-set identifiers
- CSA terms and collateral parameters
- Market data, model calibration and simulation configuration

2.2 Market risk factor simulation

Generate correlated future market states.

- Interest rates, FX rates and credit spreads initially
- Equity prices and volatility factors as future extensions

2.3 Instrument pricing

Revalue supported trades at current and simulated market states.

- Initial products: Interest Rate Swaps (IRS) and FX Forwards
- Scenario-dependent valuation across future dates

2.4 Exposure modelling

Convert simulated valuations into exposure distributions.

- EE, ENE, EPE, PFE and effective EPE
- Trade-, netting-set- and counterparty-level views

2.5 Netting and collateral

Aggregate trades under legally enforceable netting assumptions and apply collateral terms.

- Netting-set aggregation
- Variation Margin, Initial Margin, thresholds, MTA, haircuts and MPOR

2.6 XVA calculation

Calculate valuation adjustments arising from credit and funding risks.

- CVA, DVA and FVA initially
- MVA as a subsequent enhancement

2.7 Regulatory capital and RWA

Translate counterparty exposures into regulatory capital measures.

- SA-CCR: RC, PFE add-on and EAD
- Counterparty RWA and a separate CVA capital framework

2.8 Portfolio allocation

Attribute aggregate risk and capital measures to trades.

- Euler allocation
- Aumann-Shapley allocation
- Marginal contribution analysis

2.9 Reporting

Expose the outputs through interactive analytics and reproducible reports.

- Trade valuations and exposure profiles
- Netting-set risk, XVA, capital and trade attribution

3. Functional Requirements

3.1 Portfolio Input Engine

Objective: establish a standardised input interface for defining synthetic derivative portfolios.

Attribute	Description
Trade ID	Unique trade identifier
Counterparty ID	Counterparty identifier
Netting Set ID	Legal netting arrangement identifier
Product Type	IRS, FX Forward, etc.

Attribute	Description
Currency	Trade currency
Notional	Trade notional
Trade Date / Maturity Date	Trade inception and contractual maturity
Pay / Receive	Cash flow direction
Fixed Rate / Floating Index	Product-specific contractual terms
Strike / Forward Rate	Product-specific parameter

Separate configuration files should support counterparty credit information, CSA terms, market data, model parameters and simulation settings. The input engine must validate unique trade IDs, product-specific fields, dates, counterparty/netting-set mappings and market-data availability. Expected output: a validated, standardised portfolio object.

3.2 Market Simulation Engine

Objective: generate correlated future market scenarios for pricing and exposure measurement.

Risk factor	Initial model / approach
Interest rates	Hull-White One-Factor
FX rates	Geometric Brownian Motion
Credit spreads	CIR or another suitable mean-reverting model

- Calibrate models using historical or synthetic market data.
- Generate Monte Carlo paths and support multiple simulation dates.
- Model correlation between risk factors.
- Make simulation horizon, number of paths and random seed configurable.
- Return a structured scenario cube: Scenario × Simulation Date × Risk Factor.

3.3 Derivative Pricing Engine

Objective: calculate instrument valuations under current and simulated market conditions.

Product	Required capabilities
Interest Rate Swaps	Fixed-for-floating swaps; discount curve construction; forward rate estimation; projected cash flows; discounted cash flow valuation.
FX Forwards	Forward exchange rate calculation; domestic and foreign discounting; mark-to-market valuation.

- Price trades under current market conditions and reprice under simulated scenarios.
- Support multiple currencies and curves.
- Use modular product-specific pricing implementations.
- Validate against independent benchmark valuations.
- Return a valuation cube: Trade × Scenario × Simulation Date.

3.4 Exposure Modelling Engine

Objective: convert simulated trade valuations into counterparty exposure profiles.

Metric	Definition / requirement
Positive Exposure	Maximum of portfolio value and zero.
Negative Exposure	Maximum of negative portfolio value and zero.
EE	Average positive exposure at a future date.
ENE	Average negative exposure at a future date.
EPE	Time-weighted or averaged EE over the exposure horizon.
PFE	Selected percentile of the exposure distribution.
Effective EPE	Regulatory exposure measure based on effective EE.

- Configure PFE confidence levels and exposure horizons.
- Support trade-level and portfolio-level exposure distributions.
- Visualise exposure profiles over time.
- Compare standalone, gross and netted exposure.

3.5 Netting and Collateral Engine

Objective: calculate economically relevant exposure after recognising enforceable netting and collateral arrangements.

Netting requirements

- Group trades by netting-set identifier.
- Aggregate positive and negative trade valuations within each netting set.
- Calculate netting-set exposure and compare gross versus net exposure.
- Support multiple netting sets for the same counterparty.

Collateral requirements

Parameter	Description
Threshold	Uncollateralised exposure threshold
Minimum Transfer Amount	Minimum collateral transfer
Independent Amount	Additional contractual collateral
Variation Margin	Periodic mark-to-market collateral
Initial Margin	Additional risk collateral
Haircut	Collateral valuation adjustment
MPOR	Margin period of risk
Margin Frequency	Daily, weekly, etc.

- Simulate collateral balances and margin calls over time.
- Implement call and settlement logic, MPOR, collateral currencies and haircuts.
- Distinguish collateralised and uncollateralised exposure.
- Produce netting-set-level collateral-adjusted exposure profiles.

Core conceptual calculation: netting-set exposure = max(sum of trade MTMs, 0), before collateral and other applicable adjustments.

3.6 XVA Calculation Engine

Objective: quantify valuation adjustments arising from counterparty credit risk, own credit risk and funding requirements.

CVA

- Implement unilateral CVA initially, followed by bilateral CVA.
- Model counterparty hazard rates, survival probabilities and marginal default probabilities.
- Incorporate LGD, discounted expected exposure and integration across future exposure periods.

DVA and FVA

- Calculate DVA using negative exposure profiles and own-credit assumptions.
- Implement a simplified FVA using expected funding requirements and funding spreads.

MVA and outputs

- Introduce MVA subsequently using initial margin requirements and funding costs.
- Report XVA at trade level where meaningful, netting-set, counterparty and portfolio levels.
- Document hedging, funding and close-out conventions; do not assume all adjustments are mechanically additive.

3.7 Capital and RWA Engine

Objective: translate counterparty exposures into regulatory capital requirements.

SA-CCR

- Implement Replacement Cost (RC) and Potential Future Exposure (PFE) add-on.
- Apply relevant multiplier and collateral adjustments.
- Calculate Exposure at Default (EAD) and RWA.
- Initially support interest rate and FX asset classes; add credit, equity and securities financing products later.

Capital outputs

Metric	Aggregation level
RC	Netting set
PFE add-on	Netting set
EAD	Netting set
Risk weight	Counterparty
RWA	Counterparty / portfolio
CVA capital	Counterparty / portfolio

CVA capital should be implemented as a separate module supporting a selected, documented regulatory methodology. SA-CCR capital and CVA capital must remain distinct calculations. The initial release is an analytical implementation of a selected framework, not a claim of regulatory production approval.

3.8 Risk and Capital Allocation Engine

Objective: attribute aggregate risk and capital measures to individual trades. This is a core differentiating component of Quant Lab.

Euler allocation

For a differentiable, positively homogeneous portfolio risk measure R, the Euler contribution of trade i can be expressed as:

$$RC_i = V_i \times (\partial R / \partial V_i)$$

where V_i is the relevant trade-level scaling variable. Under the applicable mathematical assumptions, the sum of trade contributions reconciles to the portfolio measure. Potential applications include exposure, CVA, EAD and RWA allocation, subject to each measure's properties.

Aumann-Shapley and marginal contribution

- Implement Aumann-Shapley allocation for suitable nonlinear portfolio measures.
- Calculate changes in portfolio metrics when trades are added or removed.
- Reconcile allocations to aggregate metrics where the methodology supports reconciliation.
- Handle zero and negative contributions and identify numerical instability.
- Support numerical differentiation and finite-difference benchmarking.

Illustrative output

Trade	Standalone EAD	Allocated EAD	Allocated RWA	Allocated CVA
Trade A	…	…	…	…
Trade B	…	…	…	…
Trade C	…	…	…	…
Portfolio	—	Total	Total	Total

Allocation methods must be selected according to the mathematical properties of each measure. Euler allocation must not be forced onto a metric that does not satisfy its assumptions.

4. Non-Functional Requirements

Area	Requirement
Language	Python
Architecture	Modular, object-oriented
Data handling	Pandas / NumPy
Numerical computing	SciPy
Simulation	NumPy / SciPy
Validation	Pytest
Configuration	YAML / JSON
Data storage	Parquet / SQLite
Visualisation	Plotly
Code quality	Type hints, docstrings and linting
Version control	Git
Reproducibility	Deterministic configuration and random seeds
Performance	Vectorised calculations and parallel simulation where appropriate

Engineering requirements

- Separate business logic, quantitative models and application interfaces.
- Use schema-based validation for inputs and outputs.

- Implement consistent exception handling and logging.
- Unit-test individual models and integration-test the complete workflow.
- Benchmark computationally intensive modules.
- Version model configurations and assumptions.

5. Proposed Software Architecture

The platform should follow a layered architecture, with each layer exposing well-defined interfaces.

Layer	Components
1. Data Layer	Trade data; counterparty data; CSA data; market data
2. Core Quantitative Layer	Market simulation; curve construction; pricing models; exposure modelling
3. Portfolio Risk Layer	Netting engine; collateral engine; exposure aggregation
4. Valuation Adjustment Layer	CVA; DVA; FVA; MVA
5. Capital Layer	SA-CCR; CVA capital; RWA
6. Allocation Layer	Euler allocation; Aumann-Shapley; marginal contributions
7. Application Layer	Input interface; configuration; execution orchestration
8. Reporting Layer	Exposure dashboard; XVA dashboard; capital dashboard; trade attribution

Individual engines should be executable independently and through a single orchestrated workflow. The application should preserve intermediate outputs to support debugging, reproducibility and traceability.

6. Application Interface and Reporting

The initial application should provide a lightweight user interface. A Streamlit-based interface is sufficient for the first release.

Dashboard	Key content
Portfolio Overview	Trade, counterparty and netting-set counts; portfolio notional; aggregate MTM.
Exposure Analytics	EE and PFE curves; gross versus net exposure; collateralised versus uncollateralised exposure; netting benefit.
XVA Analytics	CVA, DVA and FVA; counterparty breakdown; exposure and default probability profiles.
Capital Analytics	RC, PFE add-on, EAD, RWA and CVA capital.
Allocation Analytics	Trade-level Euler contributions; RWA attribution; CVA contributions; standalone versus allocated metrics.

7. Implementation Roadmap

Phase	Scope	Deliverable
1	Data architecture and trade ingestion	Portfolio input engine
2	Market simulation	Monte Carlo risk factor engine
3	Pricing	IRS and FX Forward pricing
4	Exposure modelling	EE, EPE and PFE
5	Netting and collateral	Netting-set exposure engine
6	XVA	CVA, DVA and FVA
7	Capital	SA-CCR and RWA
8	Allocation	Euler and Aumann-Shapley
9	Application	Integrated dashboard
10	Optimisation	Performance, testing and documentation

Delivery increments

- **MVP — Phases 1–5:** a working end-to-end exposure modelling platform.
- **Version 2 — Phases 6–8:** valuation adjustments, regulatory capital and allocation capabilities.
- **Version 3 — Phases 9–10:** a polished application with performance optimisation and documentation.

8. Acceptance Criteria

ID	Acceptance criterion
AC01	Synthetic portfolios can be ingested and validated.
AC02	Market risk factors can be simulated using calibrated models.
AC03	Supported derivatives can be priced under simulated market conditions.
AC04	Trade-level future exposure profiles can be generated.
AC05	Exposures can be aggregated at netting-set level.
AC06	Collateral agreements affect exposure calculations correctly.
AC07	CVA, DVA and FVA can be calculated.
AC08	SA-CCR EAD and RWA can be calculated.
AC09	Euler allocation reconciles to aggregate portfolio metrics wherever applicable.
AC10	Aumann-Shapley allocation can be compared against alternative methods.
AC11	Key outputs can be traced to source trades and model assumptions.
AC12	The complete workflow can be executed through a single application interface.

9. Quantitative Validation Framework

Quant Lab should not merely produce numbers. It must demonstrate that calculations are mathematically and financially consistent.

Area	Mandatory validation checks
Pricing	Compare IRS valuations with independently calculated discounted cash flows; compare FX Forward valuations with analytical formulas; test rate and FX sensitivities.
Simulation	Compare simulated means and variances with theoretical expectations; check correlation matrices; validate interest-rate mean reversion.
Exposure	Ensure positive exposure is never negative; test netting benefits under appropriate assumptions; validate PFE against scenario distributions.
Collateral	Reconcile collateral movements to thresholds and transfer rules; confirm collateral reduces exposure under applicable assumptions.
XVA	Check CVA sensitivity to PD, LGD and exposure; verify discounting and default-probability integration; check non-negativity under standard unilateral assumptions.
Capital	Reconcile SA-CCR components against independent benchmarks; validate EAD and RWA aggregation.
Allocation	Test allocation reconciliation; compare Euler contributions with finite-difference marginal contributions; examine behaviour under portfolio scaling.

10. Future Enhancements

The following capabilities are intentionally excluded from the initial implementation but should be considered in the architecture.

- **Advanced pricing:** Hull-White multi-factor, SABR and Heston models.
- **Wrong-Way Risk:** dependence between counterparty credit quality and exposure.
- **Multi-Curve Framework:** OIS discounting and tenor-specific forwarding curves.
- **Dynamic Initial Margin:** SIMM-inspired sensitivities and margin calculations.
- **Advanced Collateral:** multiple collateral currencies, cheapest-to-deliver optimisation and collateral optionality.
- **Advanced Capital:** IMM-style exposure methodology and additional CVA capital methodologies.
- **Performance Engineering:** GPU acceleration, multiprocessing and distributed Monte Carlo.
- **Model Risk:** calibration monitoring, model benchmarking and sensitivity analysis.
- **Portfolio Optimisation:** trade-level capital efficiency and counterparty concentration analysis.

11. Definition of Success

Quant Lab will be successful when it demonstrates the ability to build an integrated quantitative risk platform rather than simply implement individual financial formulas.

- Translate financial contracts into computational representations.
- Build stochastic market models and pricing engines.
- Measure exposure under simulated future market conditions.
- Understand the interaction between netting, collateral and counterparty risk.
- Calculate valuation adjustments and regulatory capital.
- Apply quantitative allocation methodologies to nonlinear portfolio measures.
- Design scalable, testable and maintainable quantitative software.

ULTIMATE DELIVERABLE

A reproducible, end-to-end derivatives counterparty risk platform that connects trade-level economics to portfolio-level valuation, exposure, capital and risk attribution.

Document note: This PRD defines the intended functional scope and initial implementation direction. Detailed quantitative conventions, regulatory interpretations, model assumptions and technical interfaces should be specified in the subsequent technical design document.